DEPARTMENT EXPECTATIONS

App Development

Led by **Rishi**

Bring an idea to a phone. Explore iOS, Flutter, or Android.

Shared basics

- Object-oriented programming: classes, objects, and inheritance
- Simple problem solving: reversing a string, summing a list, and factorial logic
- MVVM and MVC architecture

Tracks

Explore one of these three tracks alongside the shared basics.

Swift · iOS

- Swift syntax, variables, optionals, and functions
- Struct versus class, @State, and @Binding
- SwiftUI: VStack, HStack, ZStack, Text, Button, and Image
- Basic state management, such as a counter app
- Navigation and NavigationStack basics

Flutter · Dart

- Dart variables, functions, lists, maps, and null safety
- Stateless and Stateful Widgets
- Column, Row, Container, Text, Image, and Scaffold
- setState(), widget rebuilds, and the widget tree
- MaterialApp → Scaffold → Body

Kotlin · Android

- Kotlin syntax, val versus var, functions, and null safety (?, !!)
- Android Studio, activities, MainActivity, and XML layouts
- findViewById and setOnClickListener
- Intents and navigating between activities
- TextView, Button, and EditText

Submitting an application does not confirm selection. Shortlisting and an offline round follow, and their date, time, and venue will be announced.